

Mathpix API Demo

How to measure distance between two points?

The **distance formula** is used to calculate the distance between two points (x_1, y_1) and (x_2, y_2) in a **2D plane**. It is derived from the **Pythagorean theorem**.

Formula:

THE DISTANCE FORMULA

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

What is Permutation?

A **permutation** is an arrangement of objects in a specific order. It represents the different ways of ordering a given set of elements.

Formula:

Permutation

$${}^n\mathbf{P}_r = \frac{n!}{(n-r)!}$$

where,

- **n = Number of Total Objects**
- **r = Number of Objects Chosen at Once**
- **$0 \leq r \leq n$**